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Fake Review Detection Using TF-IDF + Logistic Regression

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Project

Fake Review Detection



OUTLINE

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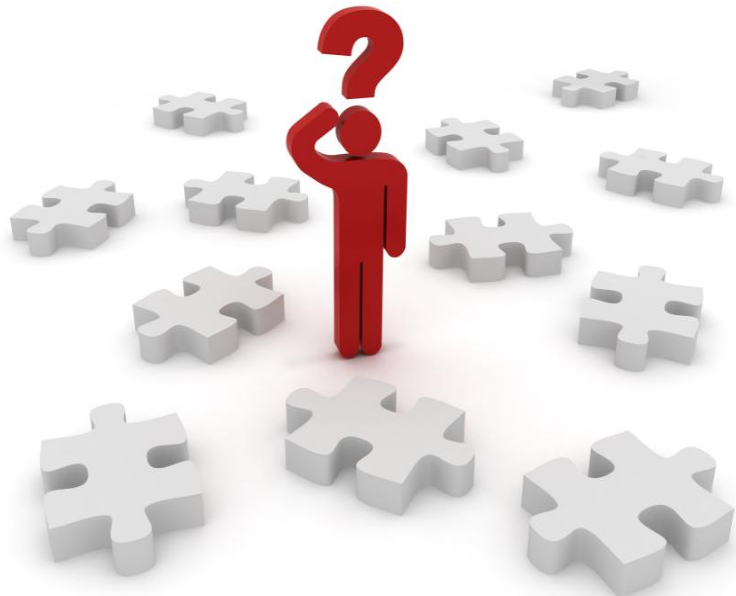


Introduction

- Online product reviews strongly influence purchase decisions.
- Some reviews may be misleading, manipulated or artificially generated.
- Manual verification of thousands of reviews is difficult.
- The project develops an automated machine-learning based classification system.
- The system uses review text and TF-IDF features with Logistic Regression.

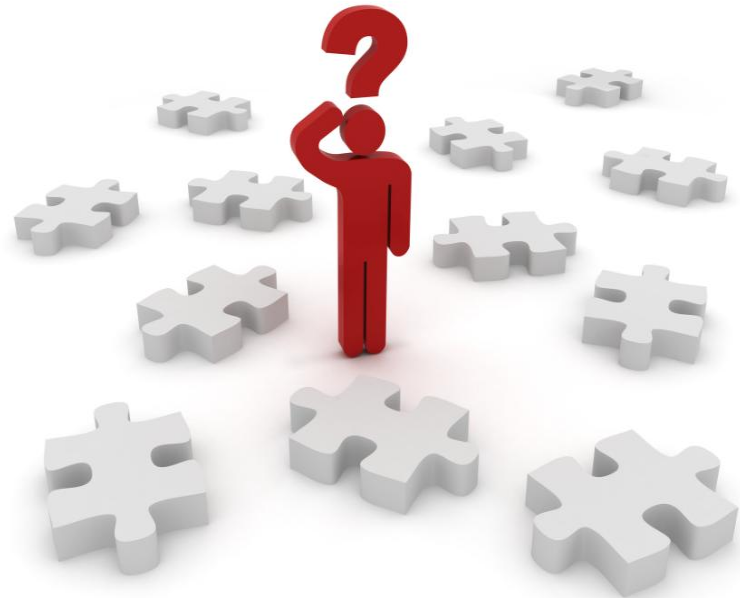
Dataset Overview

- Total Reviews: 40,432
- Columns: category, rating, label and text_
- Categories: 10
- Target Classes: CG and OR
- Ratings range from 1 to 5
- Review text is the main input for NLP classification.



Problem Statement

- Fake or unreliable reviews can distort consumer decisions.
- Large review collections make manual screening expensive and slow.
- Text contains patterns that can be learned by supervised ML models.
- An automated classifier can provide a fast preliminary prediction.



Objectives

- **Develop a machine-learning system for review classification.**
- **Clean and normalize review text.**
- **Convert text into numerical features using TF-IDF.**
- **Train and evaluate a Logistic Regression model.**
- **Provide a simple prediction interface foundation.**
- **Reduce the effort required for preliminary review screening.**



Proposed Solution

- **Data Collection:** Load and understand the review dataset.
- **Text Preprocessing:** Normalize review text and remove common stop words.
- **TF-IDF:** Convert text into numerical feature vectors.
- **Train/Test Split:** Use an 80 / 20 stratified split.
- **Model Training:** Train Logistic Regression on TF-IDF features.
- **Evaluation:** Measure Accuracy, Precision, Recall, F1-Score and Confusion Matrix.



System Architecture

User Input

- User Review
- Review Text
- Rating
- Category

Machine Learning Model

- TF-IDF feature extraction
- Logistic Regression training
- Model evaluation
- Classification

Data Preprocessing

- Lowercase text
- Remove stop words
- Tokenize text
- Prepare clean review text

Career Recommendation

- Predicted result
- CG – Computer Generated
- OR – Original / Real
- Confidence score

Live Demo of Project

- Here's the list of some demo steps –
- Enter a product review in the prediction interface
- Click the Predict Review button
- View the predicted class and confidence
- Test both original and computer-generated review examples



Machine Learning Model

Model trained successfully!

Model Accuracy: 86.51% Training Method: TF-IDF + Logistic Regression

Classification Report

	precision	recall	f1-score	support
CS	0.875	0.885	0.880	3022
CR	0.858	0.875	0.866	3032
accuracy	0.865	0.865	0.865	6055
macro avg	0.865	0.865	0.865	6055
weighted avg	0.865	0.865	0.865	6055

Predict a New Review

Enter a product review:

10

Predict Review

Prediction: COMPUTER GENERATED / FAKE REVIEW

Predict a New Review

Enter a product review:

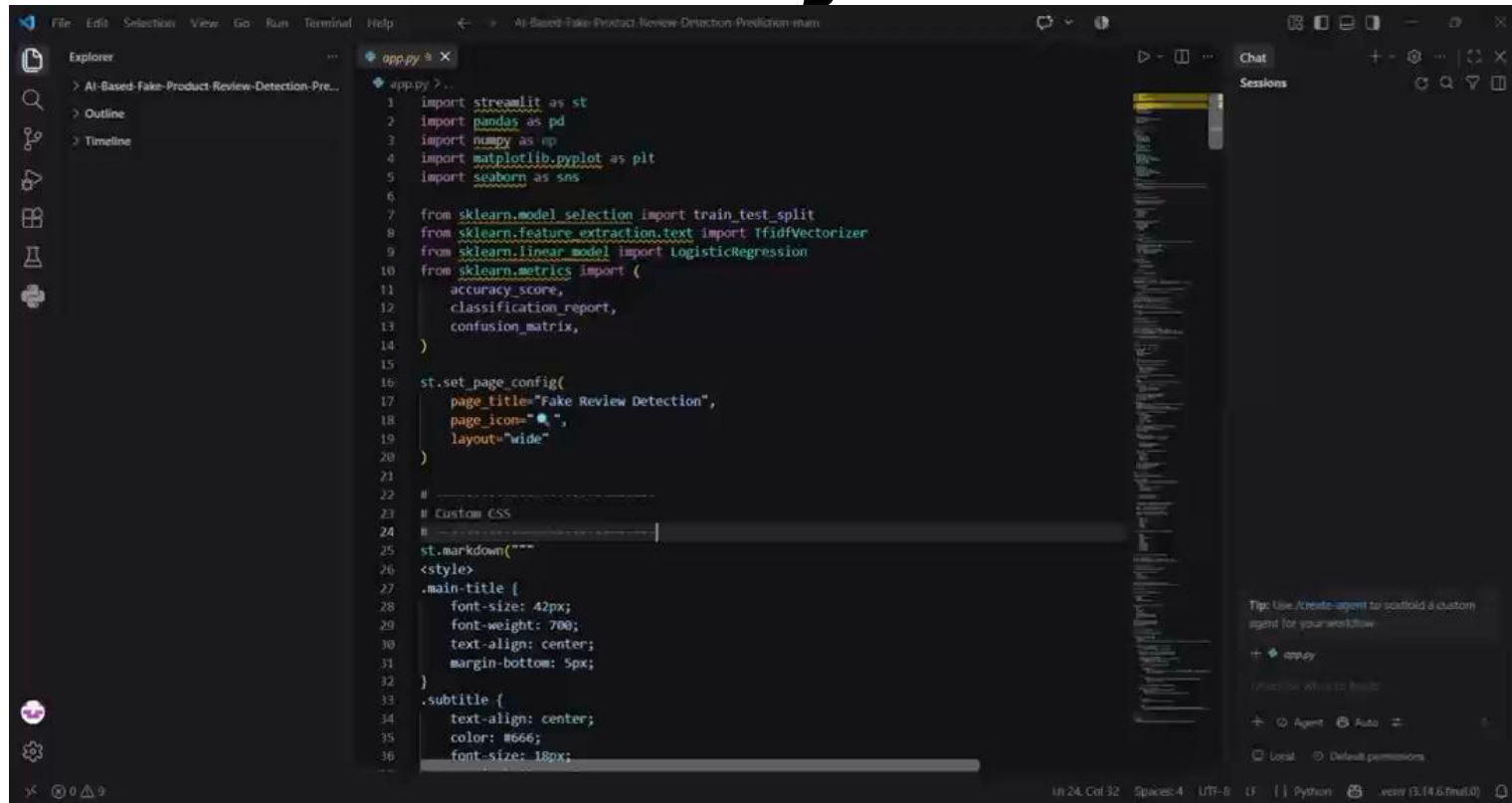
200

Predict Review

Prediction: ORIGINAL / REAL REVIEW

Confidence: 93.45%

Video of Project Demo



```
File Edit Selection View Go Run Terminal Help
AI-Based-Fake-Product-Review-Detection-Prediction-main
app.py X
1 import streamlit as st
2 import pandas as pd
3 import numpy as np
4 import matplotlib.pyplot as plt
5 import seaborn as sns
6
7 from sklearn.model_selection import train_test_split
8 from sklearn.feature_extraction.text import TfidfVectorizer
9 from sklearn.linear_model import LogisticRegression
10 from sklearn.metrics import (
11     accuracy_score,
12     classification_report,
13     confusion_matrix,
14 )
15
16 st.set_page_config(
17     page_title="Fake Review Detection",
18     page_icon="📊",
19     layout="wide"
20 )
21
22 # -----
23 # Custom CSS
24 # -----
25 st.markdown("""
26 <style>
27 .main-title {
28     font-size: 42px;
29     font-weight: 700;
30     text-align: center;
31     margin-bottom: 5px;
32 }
33 .subtitle {
34     text-align: center;
35     color: #666;
36     font-size: 18px;
```

Chat

Sessions

Tip: Use /create-agent to scaffold a custom agent for your workflow

+ create-agent

Python Agent Auto

Local Default permissions

Ln 24, Col 32 Spaces: 4 UTF-8 LF Python .env (3,14.6,final.0)

Conclusion

- The project combines NLP preprocessing, TF-IDF and Logistic Regression.
- The supplied dataset provides a balanced two-class review classification task.
- The reproducible evaluation achieved 86.46% accuracy on held-out data.
- The workflow can classify reviews as CG or OR and can be extended into a simple Streamlit-based application.



Future Scope

- Use larger and more diverse datasets.
- Compare additional ML/NLP models.
- Add multilingual detection.
- Integrate external fact-checking signals.
- Add confidence scores and error analysis.



Technology & Future Scope

- Python, Pandas and NumPy
- Scikit-learn for TF-IDF and Logistic Regression
- Jupyter / VS Code / Google Colab
- Streamlit for optional interactive deployment
- Improve the classifier with better data and additional models
- Test performance on unseen domains



Thank You